

MD TANVIR ISLAM

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EDUCATION AND AWARDS

March 2023 June 2025	M.Sc., SUNGKYUNKWAN UNIVERSITY (SKKU), South Korea <i>Department of Computer Science and Engineering</i> CGPA : 4.19/4.50 Thesis : Study Towards Robust Image Enhancement in Adverse Atmospheric Conditions.	Transcript
March 2022 February 2023	Korean Language and Cultural Program, PAI CHAI UNIVERSITY, South Korea <i>Korean Language and Cultural Program</i> TOPIK Level : 4/6 As a prerequisite, I completed this program under the Global Korea Scholarship (GKS).	Transcript
January 2015 January 2019	B.Sc., North Western University (NWU), Bangladesh <i>Department of Computer Science and Engineering</i> CGPA : 3.98/4.00 Thesis : Study on diabetes prediction for typical and non-typical cases using machine learning approaches.	Transcript

RESEARCH EXPERIENCES

IF : IMPACT FACTOR

March 2023 Present	Research Assistant, VIS2KNOW LAB , Sungkyunkwan University, South Korea ➤ Data curation and research in AI image detection, image dehazing, and low-light image enhancement. ➤ Our research on image dehazing, LLIE, and AI-synthesized image detection has led to three papers are published at top-tier conferences and journals such as <i>ACM Multimedia</i> (Rank A), <i>Asian Conference on Computer Vision (ACCV)</i> (Rank B), <i>Alexandria Engineering Journal (AEJ)</i> (IF : 6.2, Q1). Also, three of our innovations are registered at <i>US Patent Office</i> .	Lab Director
July 2018 August 2021	Research Assistant, CSHI-RLAB , North Western University, Bangladesh ➤ Collected real-life datasets to perform research using machine-learning techniques and published several papers at different international conferences, which achieved around 100 citations so far.	Lab Director

TEACHING EXPERIENCES

March 2024 August 2024	Teaching Assistant, DEPARTMENT OF AI, Sungkyunkwan University, South Korea ➤ Assisted with curriculum planning for courses and held office hours to support and mentor students. ➤ Facilitated group discussions and study sessions to enhance collaborative learning.
January 2019 December 2020	Part-time Lecturer, SOUTH-EAST ENGINEERING COLLEGE, Khulna, Bangladesh ➤ Planned, prepared and presented lectures while assessing student assignments. ➤ Maintained communication with teachers and students.

AWARDS AND SCHOLARSHIPS

November 2024	Academic Excellence Award Winner, NIIED, Government of South Korea The NIIED selected me as one of the three students for this award based on the research outcomes.	Certificate
December 2023	16th Korean Language Society Scholarship, PAI CHAI UNIVERSITY, South Korea Received this award based on my academic performance and on-campus activities including volunteering.	Certificate
August 2022 February 2025	Global Korea Scholarship (GKS), NIID, South Korea Received this award based on my academic performance and on-campus activities including volunteering.	Certificate
January 2015 January 2019	First Position in Merit List, NORTH WESTERN UNIVERSITY, Bangladesh Certified by my undergraduate university for securing the 1 st position in the merit list among all students.	Certificate

- SCIE | Q1 | IF : 6.2** **Md Tanvir Islam**, Ik Hyun Lee, Ahmed Ibrahim Alzahrani, and Khan Muhammad*. "MEXFIC : A meta ensemble eXplainable approach for AI-synthesized fake image classification." [Alexandria Engineering Journal](#) 116 (2025) : 351-363. (Published)
- SCIE | Q1 | IF : 7.8** **Md Tanvir Islam**, Safkat Shahrier Swapnil, Md Masum Billal, Asif Karim*, Niussha Shafiabady, and Md Mehedi Hassan*. "Resource constraint crop damage classification using depth channel shuffling." [Engineering Applications of Artificial Intelligence](#) 144 (2025) : 110117. (Published)
- SCIE | Q1 | IF : 6.2** **Md Tanvir Islam**, Abu Saleh Musa Miah, M. Raihan, ... and Khan Muhammad*. "DiaBD : A novel benchmark dataset for diabetes prediction." [Alexandria Engineering Journal](#) 116 (2025) : 351-363. (Accepted)
- Rank A** **Md Tanvir Islam**, Nasir Rahim, Saeed Anwar, Muhammad Saqib, Sambit Bakshi, and Khan Muhammad*. "Hazespace2m : A dataset for haze aware single image dehazing." In Proceedings of the [ACM Multimedia \(ACM MM\)](#), pp. 9155-9164. 2024. (Published)
- Rank B** **Md Tanvir Islam**, Inzamamul Alam, Simon S. Woo*, Saeed Anwar, I. K. Lee, and Khan Muhammad*. "LoLI-Street : Benchmarking Low-Light Image Enhancement and Beyond." In Proceedings of the [Asian Conference on Computer Vision \(ACCV\)](#), pp. 1250-1267. 2024. (Published)
- Rank A** Inzamamul Alam, **Md Tanvir Islam**, and Simon S. Woo*. "Saliency-Aware Diffusion Reconstruction for Effective Invisible Watermark Removal." [ACM Web Conference, WWW 2025](#). (Published)
- Rank A** Inzamamul Alam, **Md Tanvir Islam**, Simon S. Woo*, Khan Muhammad. "WaveGuard : Advancing Invisible Watermarking with Spectral Projection." [International Conference on Computer Vision, ICCV 2025](#). (Accepted)
- Rank A** **Md Tanvir Islam**, Inzamamul Alam, and Simon S. Woo, Khan Muhammad*. "WeatherIR : Degradation Aware Multi-Weather Image Restoration." [Winter Conference on Applications of Computer Vision, WACV 2026](#). (In review)
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- U.S. Patent** **Md Tanvir Islam**, Ik Hyun Lee, Abdul Khader Jilani Saudagar, Abdullah Altameem, Mohammed Abaoud, Khan Muhammad. "Method for image dehazing of vehicular images." [U.S. Patent No. 12277682](#), issued April 15, 2025. (Registered)
- U.S. Patent** **Md Tanvir Islam**, Ik Hyun Lee, Abdul Khader Jilani Saudagar, Abdullah Altameem, Mohammed Abaoud, and Khan Muhammad. "Apparatus and method for single image dehazing considering haze type classification." [U.S. Patent No. 12217398](#), issued February 4, 2025. (Registered)
- U.S. Patent** Abdul Khader Jilani Saudagar, Safkat Shahrier Swapnil, Sandip Kumer Sarker, Arpon Bose Ditya, **Md Tanvir Islam**, Md. Abu-Talha Roni, Hitoun A. Alsagri, and Khan Muhammad. "Adaptive Speed Control For Line-Following Robots and Method Thereof." [U.S. Patent No. 12287650](#), issued April 29, 2025. (Registered)
- U.S. Patent** Shehzad Ali, **Md Tanvir Islam**, and Khan Muhammad. "Apparatus and method for adaptive response of autonomous vehicles in traffic intersections with flashing signals." *U.S. Patent* (In Review)
- U.S. Patent** Shehzad Ali, **Md Tanvir Islam**, and Khan Muhammad. "Apparatus and method for child hazardous activity anticipation and risk assessment system." *U.S. Patent* (In Review)
- Book Chapter** Rezuana Haque and **Md Tanvir Islam** (2024). "Neural Dynamics : Unravelling the Complexity of Brain Activity." In *Brain Networks in Neuroscience : Personalization Unveiled Via Artificial Intelligence*. Chapter 1, pp. 1-19. (In review)

SKILLS AND INTERESTS

Language English : [IELTS \(2019\) : 7.0, no band < 6.5.](#)

Korean : [TOPIK \(2022\) : 4/6.](#)

Programming Python; **Frameworks** : PyTorch, Keras, SpikingJelly; **Database** : PostgreSQL, MySQL.

Tools PowerBI, LaTeX, PyCharm, Jupyter, VSCode, Photoshop, Illustrator, Figma, Microsoft Suites.

Interests Image Restoration, Invisible Watermarking, Activity Recognition, Deepfake, Bioinformatics.

PROJECTS

HAZESPACE2M –  [GITHUB.COM/TANVIRNWU/HAZESPACE2M](https://github.com/TANVIRNWU/HAZESPACE2M)

LEAD AUTHOR

A large-scale hazy dataset of over 2 million images across 10 distinct levels and propose a novel framework for haze-aware single image dehazing that significantly improves PSNR by 2.41%, SSIM by 17.14%, and MSE by 10.2% over general dehazers.

[SOTA Image Dehazing Methods](#) [SOTA Classifiers](#) [PyTorch](#) [Overleaf](#) [PyCharm](#) [Photoshop](#)

TriFUSE & LOLI-STREET –  [GITHUB.COM/TANVIRNWU/TriFUSE](https://github.com/TANVIRNWU/TriFUSE)

LEAD AUTHOR

First to introduce a 33K paired low-light street scene dataset with object detection labels and propose TriFuse, a transformer-wavelet-diffusion model that achieves SOTA results for real-life low-light image enhancement for autonomous systems.

[LLIE Methods](#) [Diffusion Denoising](#) [Transformers](#) [Deep Learning](#) [PyTorch](#) [Overleaf](#) [PyCharm](#) [Photoshop](#)

SADRE –  [GITHUB.COM/TANVIRNWU/SADRE](https://github.com/TANVIRNWU/SADRE)

CO-AUTHOR

We introduced SADRE, which is a saliency-aware diffusion reconstruction method for effective invisible watermark removal.

[SOTA Image Watermarking Methods](#) [Attacks](#) [Diffusion](#) [PyTorch](#) [Overleaf](#) [PyCharm](#) [Illustrator](#)

TRAININGS AND LEARNING

August 2019	LICT Top-up IT Training, ICD DIVISION, Bangladesh	Certificate
September 2019	<i>Bangladesh Computer Council (Certified by George Washington University, Washington, D.C.)</i> ▶ The training includes 240 hours of technical skills such as Java, HTML, CSS, JavaScript, Bootstrap, and MVC Pattern, and 60 hours of other soft skills like presentation and communication.	
February 2018	Google Bangladesh Android StudyJam, GDG DHAKA, Bangladesh	Certificate
	<i>ICT Division, GDG Dhaka, Google Developers</i> ▶ A 225-hour Android app development course with regular mini-project assignments, culminating in a final complete app development project.	
January 2019	Self Learning,	Online, Books↓
Present	<i>Some of the courses and books I completed to grow my knowledge.</i> ▶ PyTorch for Deep Learning Bootcamp (52 hours) ▶ Python for Data Science, AI & Development (25 hours) ▶ Python for Data Science and Machine Learning Bootcamp (25 hours) ▶ Foundation of User Experience (UX) Design (19 hours) ▶ Modern Computer Vision with PyTorch ▶ Python Deep Learning Projects ▶ Advanced Deep Learning with Python	Udemy Mar 2024 IBM Mar 2024 Udemy Sept 2019 Google Oct 2021 Book by V Kishore Ayyadevara Book by Matthew Lamons Book by Ivan Vasilev

OTHER ACTIVITIES

Journal Paper Reviewer, IEEE Open Journal of the Computer Society	May 2025
Journal Paper Reviewer, Engineering Application of Artificial Intelligence	April 2025
Conference Paper Reviewer, International Conference of Pattern Recognition (ICPR) , Kolkata, India.	July 2024
Journal Paper Reviewer, PLOS One	January 2025
Journal Paper Reviewer, PLOS Digital Health	November 2024
Journal Paper Reviewer, Frontiers in Nutrition	March 2024
Founder of Animal Rescue Team, Khulna Cat Society , Khulna, Bangladesh	July 2016 - Present
Paper Presentation at ICCISIoT Conference , NIT Agartala, India	Dec 2019

“ REFERENCES

Khan Muhammad | Associate Professor

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